AX-CS SOUNDLAB GAMES USER MANUAL

OVERVIEW

The SoundLab games is made up of two applications: one for creating and one for playing choose-your-own-adventure style games, played entirely through audio narration and hand gesture or keyboard input.

This project consists of two separate applications:

1. The Game Player: for loading and playing .noui files (games) using gesture or keyboard input
2. The Game Engine: for creating, editing and saving no-gui games as .noui files

SYSTEM REQUIREMENTS

- Windows 11
- A webcam is required for gesture input detection
- A keyboard is required for keyboard input
- For AI Generation in the Game Engine, a GPU is recommended but not required. CPU-only devices are supported, the generation will just be slower.

INSTALLATION

The SounLab game is distributed as a compiled Windows executable (no Python required).

GAME PLAYER

1. Download the game player executable
2. Execute the application
3. Click ‘Ok’ to confirm registration of .noui file type on your device
4. Now, there are 2 options to play the game:
    a. Browse for a game you wish to play and click’Run’
    b. Double-click on a .noui game file. The game will be opened automatically

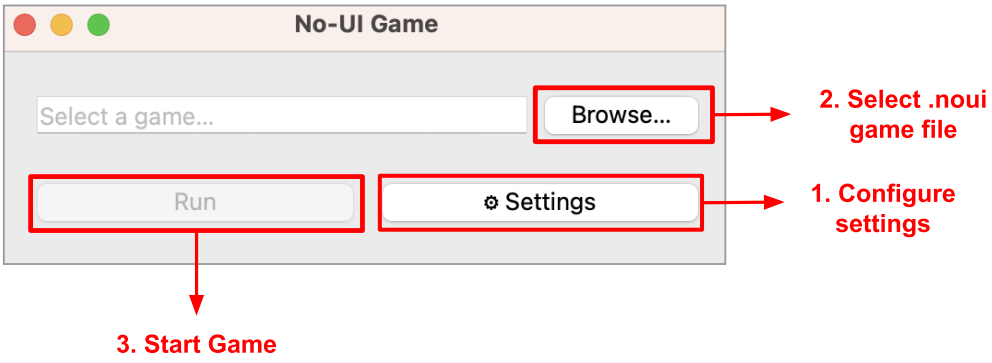
GAME ENGINE

1. Download the game engine executable
2. Execute the program

1. USING THE GAME PLAYER

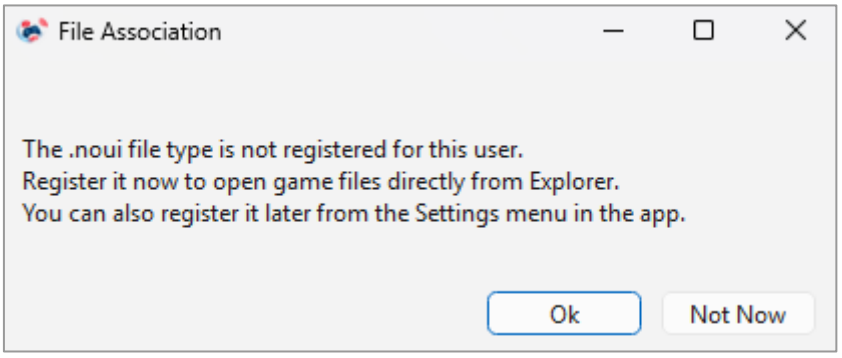
1.1 LAUNCHING GAME PLAYER

When you run gamePlayer.exe, the player page opens:

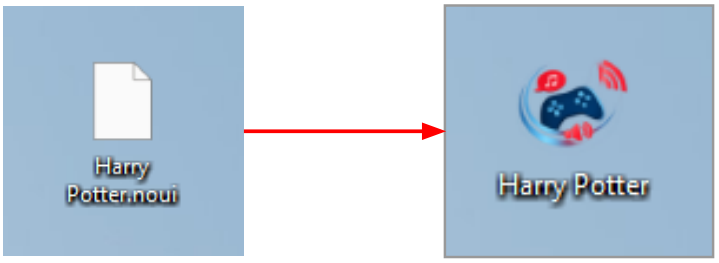


1.2 FILE TYPE REGISTRATION

Upon execution of the application, a pop-up asking the user to register .noui file type on their machine is shown. This pop-up will be shown as long as the file type is not registered.



After successful file type registration, the SoundLab games should look like this:



1.3 CONFIGURING SETTINGS

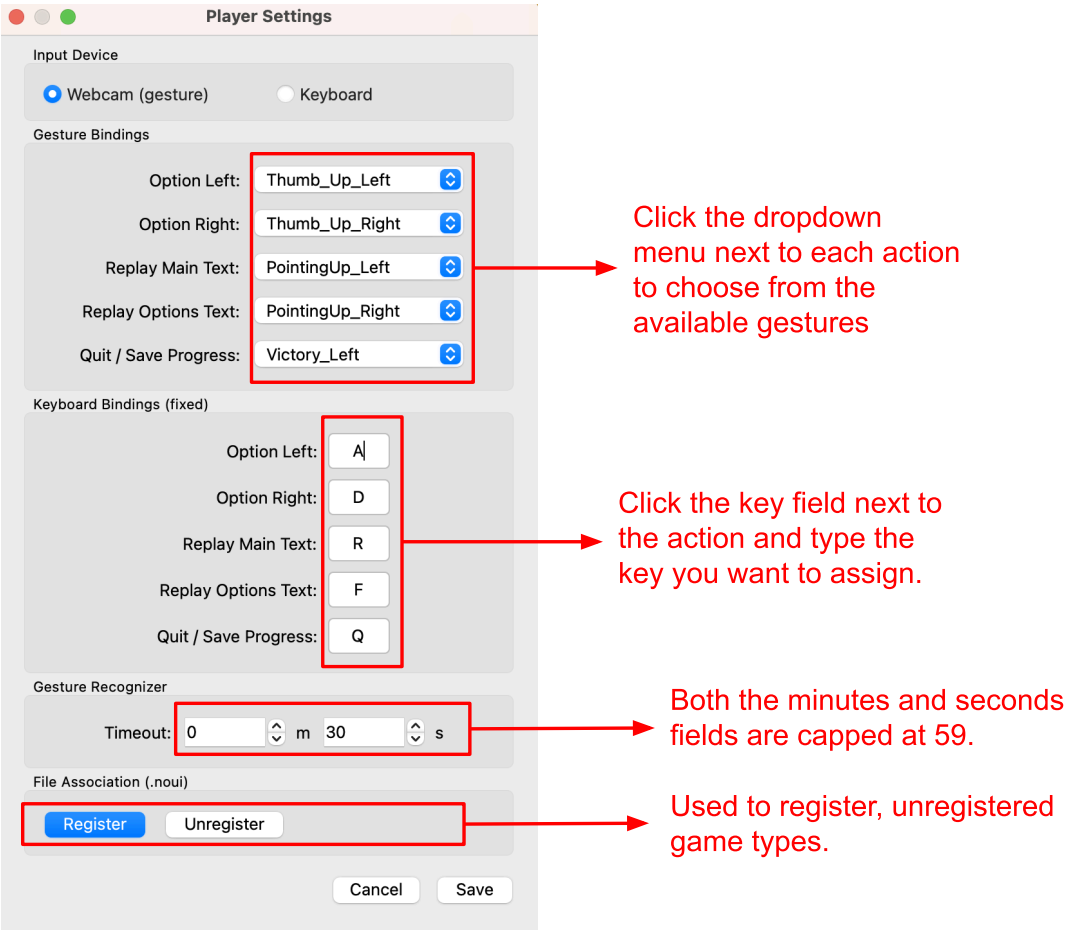
Before playing, click the Settings button to configure the game. Offered modifications are:

- **Input Device:** Select Webcam to play using hand gestures or Keyboard to play using key presses.
- **Gesture Bindings:** Each game action is mapped to a hand gesture. The available gestures can be matched to any action as long as no two actions have the same gesture.

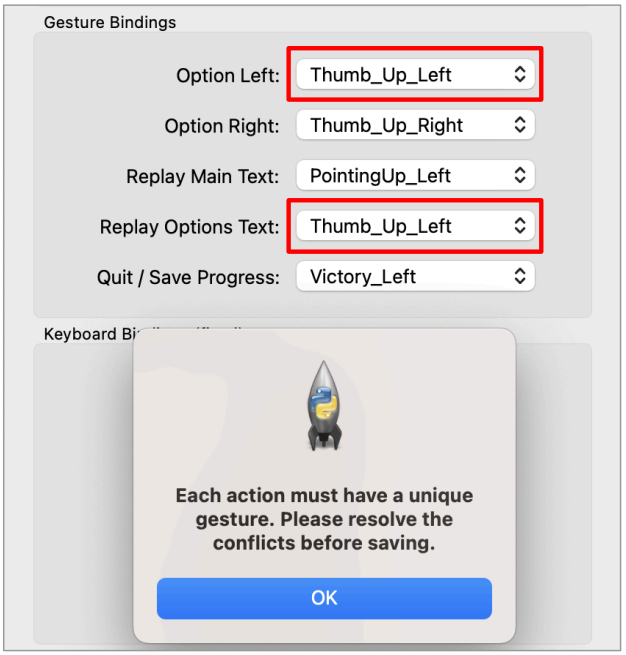
- **Keyboard Bindings:** In keyboard mode, each action is mapped to a single key. Any key can be allocated to any action but, similar to gestures, the keys must also be unique. The default bindings are:

Option Left	A
Option Right	D
Replay Main Text	R
Replay Options Text	F
Quit	Q

- **Gesture Recognizer Timeout:** The timeout sets how long a game waits for a gesture/key press before automatically saving progress and quitting due to inactivity.
- **File Association (Registeration)**



Settings are saved to settings.json and persist between sessions. Save action shows an error pop-up if the gesture or keyboard bindings are not unique.



1.4 PLAYING A GAME

After you select a .noui game using the browse button, the run button becomes unblocked. Press run to start playing the game.

The Main Menu:

When a game starts, you hear the main menu audio. This tells you the name of the game and the available options:

- Raise your right hand to start a new game
- Raise your left hand to resume from where you left off (if there is saved progress)
- Use the quit gesture to exit without playing
- Use the replay gesture to hear the main menu audio again

Game Play - Playing through a Scene:

If you do choose to play the game, from this point forward the game experience consists of scenes that prompt you to make a decision, leading to new scenes based on your choices. Each scene consists of two parts: the main narration and then the options. After both have played the game waits for your gesture:

- Raise your left or right hand to choose an option and move to the next scene
- Use the Replay Main Text gesture to hear the scene narration again
- Use the Replay Options Text gesture to hear the options again
- Use the Quit gesture to save your progress and return to main menu

Winning/Losing:

When you reach a win or lose node, the game plays scene audio followed by the win or loss message. Saved progress is cleared and the game returns to the main menu automatically.

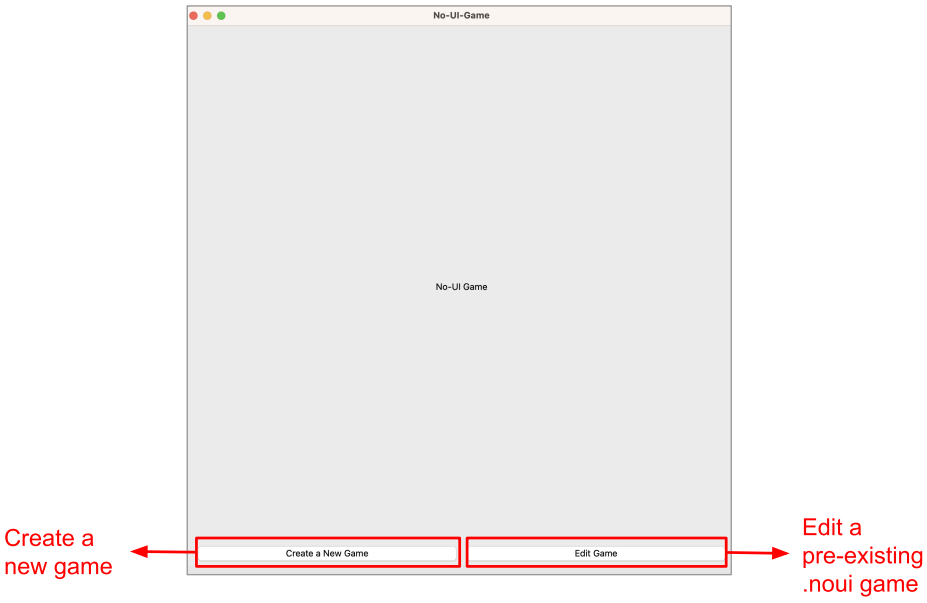
Quitting:
If you quit mid-game, your progress (the scene node you quit at) gets saved to the progress.json file in the .noui archive. The next time you run the same game you are presented with the option of resuming where you left off or starting fresh.

Inactivity Timeout:
If no gesture is detected within the timeout period defined in settings, the game plays an audio inactivity message, saves your progress and exits the main menu.

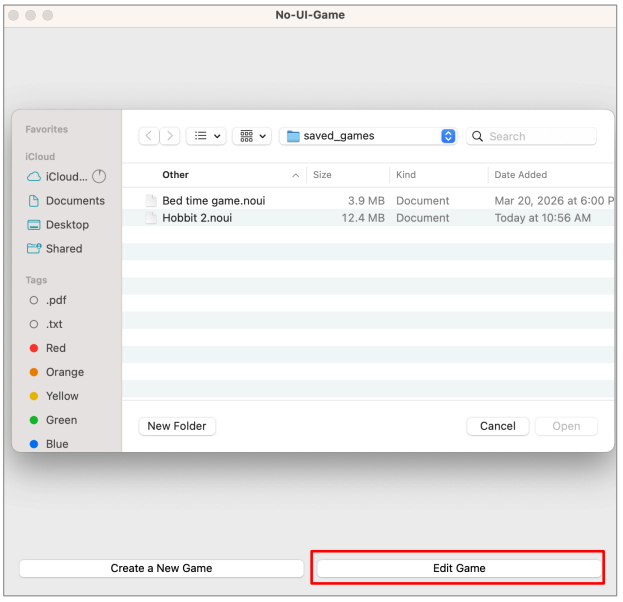
2. USING THE GAME CREATION ENGINE

2.1 LAUNCHING THE GAME ENGINE

When you run gameEngine.exe, the home page opens up:



If you press the Edit Game button it opens up a file dialog where you can choose any .noui file to edit:



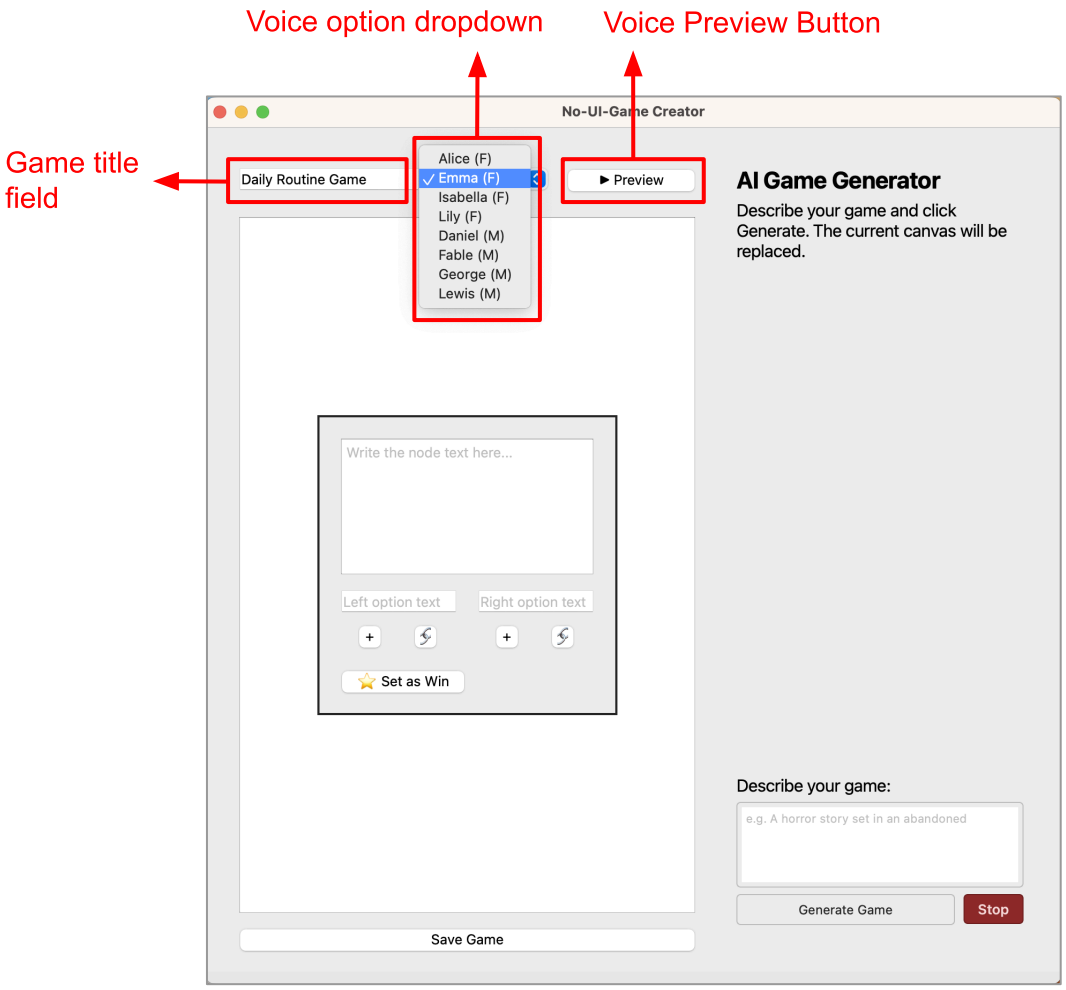
Regardless of whether you've chosen to create a game from scratch or edit a pre-existing one, the next window to open up is the game creation page. It has three main parts:

- The title bar: contains the game title field, voice selector (for narration) and a preview voice button
- The game canvas: where the game graph is built (starts of containing the root node)
- The AI helper panel: generates game graphs based on the user prompt entered.

2.2 SETTING THE GAME TITLE AND VOICE SELECTION

When creating a game you can name the game or edit the game using the text field at the top of the page.

The creator is also presented with the option of choosing the voice narrating their game. As can be seen on the dropdown menu, there are 8 British English voices available, 4 female and 4 male. And lastly by clicking the preview button, you can hear a short audio sample of the selected voice before committing to it.

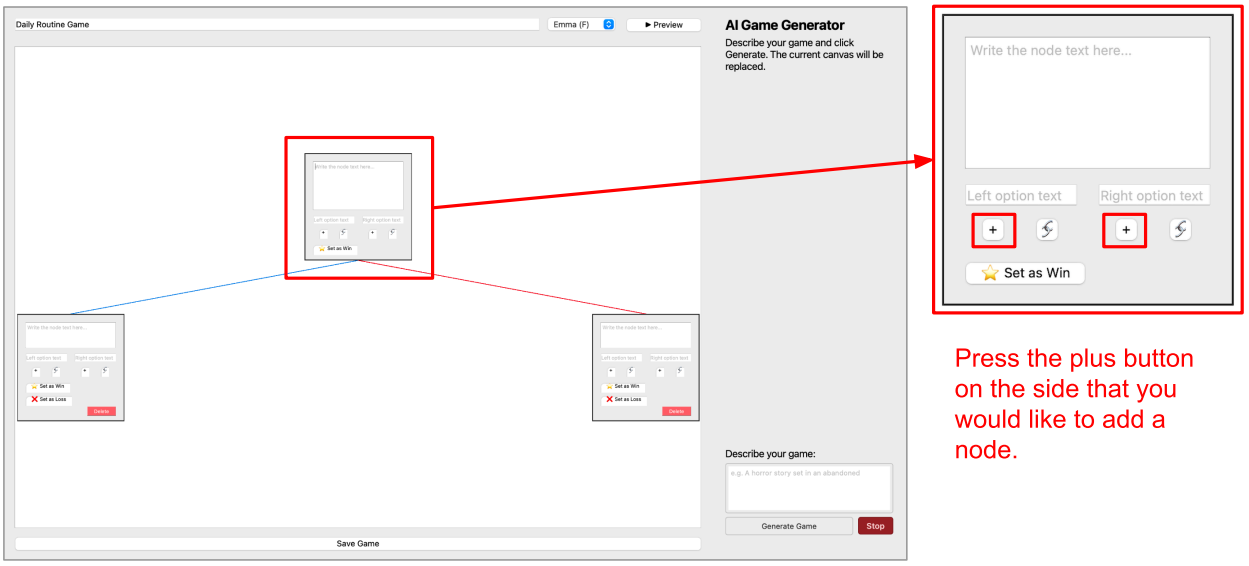


2.3 BUILDING THE STORY GRAPH

A SoundLab Game story is a branching graph where each node is a narrated story scene with (up to) two options: a left option and a right option. These options are decisions that lead to other nodes, this can either be a newly created child node to the current node or an older node that the current node links to.

Adding Nodes

To add a new node to the game graph use the plus buttons on the current nodes on the canvas. In the context of starting from scratch - with only the empty root node on the canvas, this means that you have two options of where to add a new node: as the left option result of the root node or the right option result.



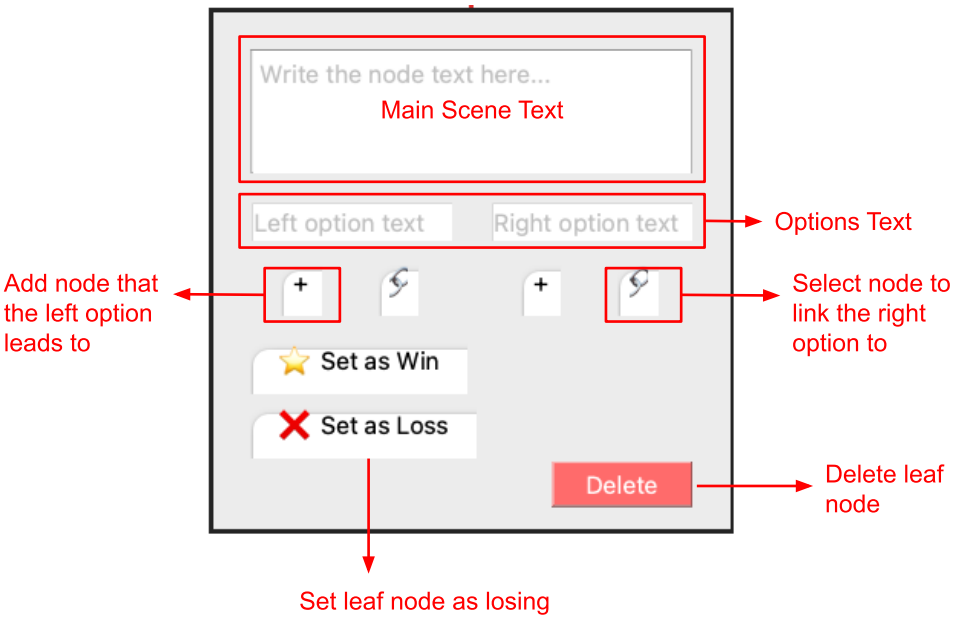
The node appears with three editable fields:

- The main scene text
- The left option text
- The right option text

In addition to that every node has an additional 5 different types of buttons:

- The plus button: adds a child node to the corresponding side
- The link button: connects the option of the corresponding side to any node on the canvas that you select
- The win button: sets the node as win
- The lose button: sets the node as loss
- The delete button: deletes the node

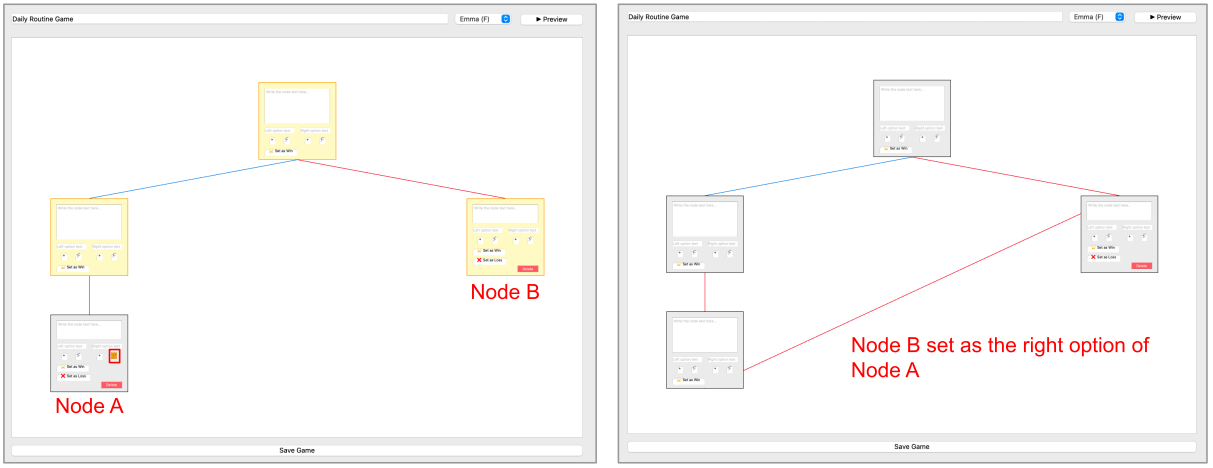
The last two mentioned are only visible on nodes that do not have options leading to other nodes (leaf nodes).



Linking Nodes

If a node option doesn't lead to its direct child, it can be linked to any other node already on the canvas with the Link Button (shown above).

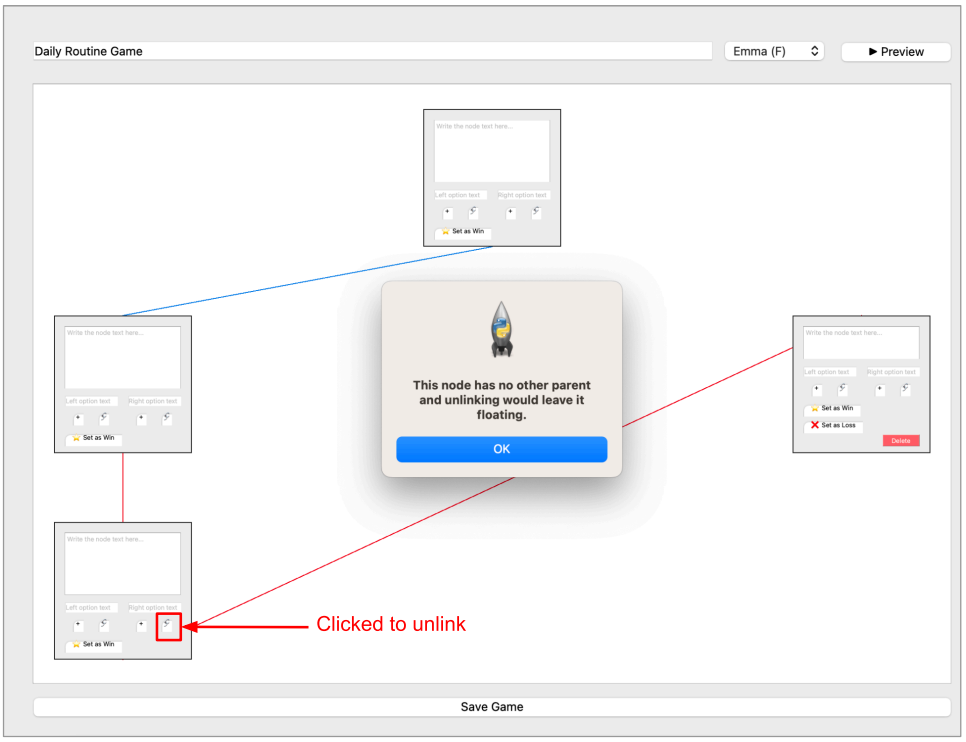
Once the button is pressed the canvas enters linking mode: all nodes available to be linked are highlighted yellow. Click on the target node to make the connection and leave linking mode.



A red line indicates a right connection, a blue line indicates a left connection.

To disconnect a linked node, click the corresponding link button.

The disconnect will be blocked by an error message if it leads to the disconnected node becoming disconnected completely from the graph (floating node).



In this case the link can be broken by deleting the target node. More on deletion later.

Setting Win and Lose Nodes

To mark a node as winning click the Set as Win button - it will be highlighted yellow.
To mark a node as losing click the Set as Loss button - it will be highlighted red.

A node cannot be set as both winning and losing, for example if the Set as Loss button is clicked on a node already set as winning, the Set as Win button will be deselected before its set as loss, and vice versa.

Deleting Nodes

Delete buttons only appear on leaf nodes as you cannot delete a node that leads to other nodes.
To delete a leaf node, simply press the Delete button.
To delete a node with children, first delete or reconnect its child nodes (nodes its options lead to).

Navigating the Game Creation Canvas

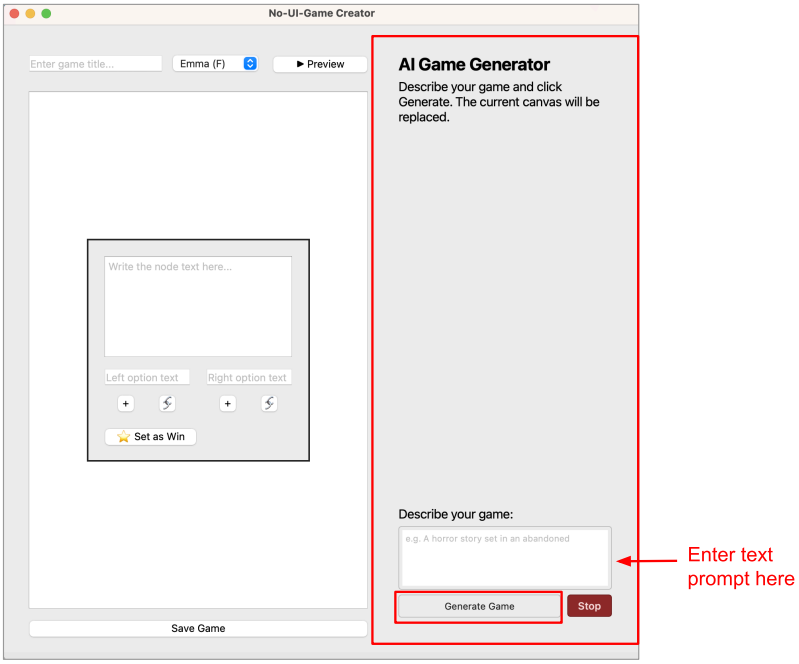
The game canvas is scrollable and has zoom in/out functionality to make looking through a large game graph easier.

- Scroll to zoom in and out

- Click and drag to move around the canvas space

2.4 AI GAME GENERATION

The AI generation panel is to the right side of the editor and can generate a complete game graph from an entered text prompt using a local LLM.



After the prompt is entered, click generate to start the generation of your SoundLab game. The editor will first generate a blueprint, the empty structure of the game graph with placeholder nodes, and then slowly populate the nodes one by one as the LLM generates the story.

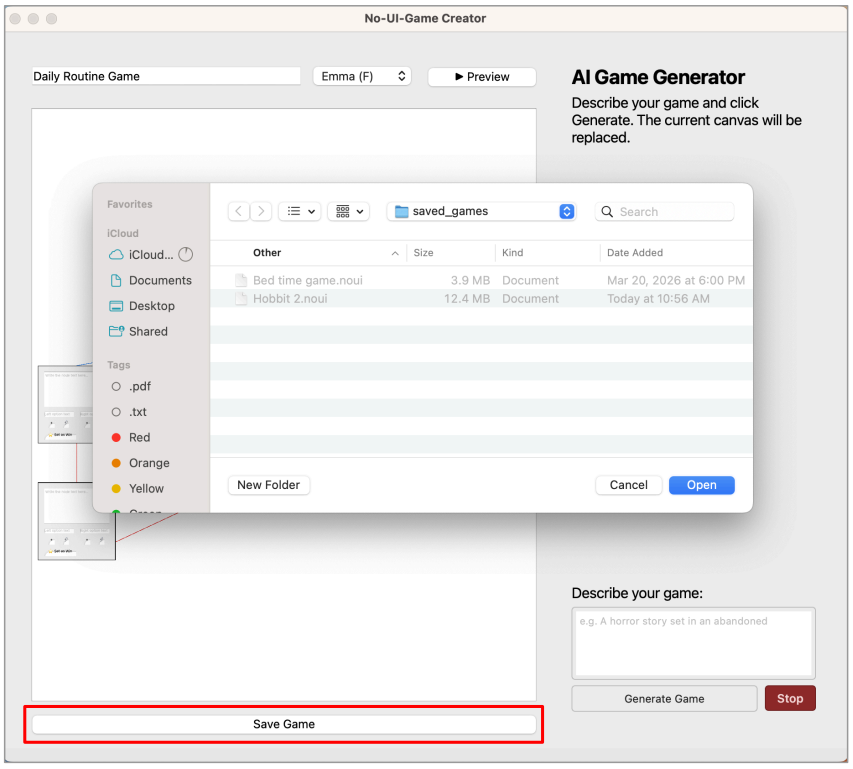
Generation can be stopped by clicking Stop at any point during generation.

Once generation is complete, the full game is shown on the canvas. And a Generation Complete text shows up below the buttons in the AI helper sidepanel. You can edit any node manually to refine the story before saving.

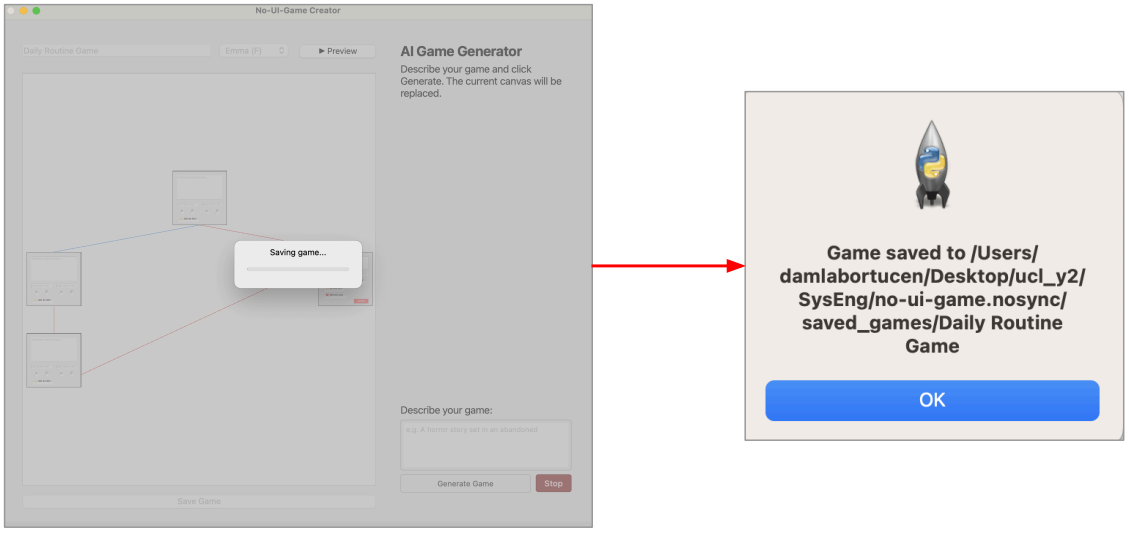
2.5 SAVING A GAME

Now that you have either written your game by hand, generated it from a prompt using the AI helper or have finished editing a pre-existing game, you can save your game.

To save a game click the save button at the bottom of the page. This will open a file dialog asking you to choose where you would like your .noui game to be saved.



The saving process serailises the graph, generates TTS audio for all nodes and system prompts using the selected voice and packages everything into a .noui file.
If the game has no title it will be saved as untitled.noui.
A success message appears when saving is completed.



TROUBLESHOOTING

Webcam not Detected

- Make sure your webcam is not being used by another application.
- Try switching to keyboard mode to make sure everything else is working correctly
- Make sure the application has webcam access in Settings > Privacy > Camera

Gesture not Recognised

- Make sure your hands are clearly visible in the webcam frame and well lit
- Try holding the gesture for a bit longer